outokumpu
high performance stainless steel

CERTIFICATE - ZEUGNIS - CERTIFICAT

EN 10204 - 3.1
25060939-EN-0

Invoice No. / Rechnung Nr. / N° de facture:
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Business Unit / QCM: BA Quarto Plate,Quarto Plate Europe
Date Datum Date: 23-Jun-2025
Load, Ladung, Charge No: 261437
Acknowledged ID, Bestätigung, Commande ID: 360027368

Your ref, Ihre Ref., Votre ref:
0236979-HOU

Buyer, Besteller, Acheteur:
Rolled Alloys, Inc.
6630 Monclova Road
43537, Maumee, Ohio
UNITED STATES OF AMERICA

Requirements, Anforderungen, Exigences:
DOW G8S-4001-51 17-NOV-2020
ASME SA-240M Code Sect. II. Part A:2023
NACE MR0103 / ISO 17945:2015
ASTM A240M / NACE MR0175/ ISO 15156-3:24/2020
Additional requirement, see "Other information"
EN 10163-2 B3

ACCEPTED
BY JL DATE 12/18/25

Consignee, Empfänger, Lieu de livraison:
Rolled Alloys

Mark of Manufacturer / Zeichen des Lieferwerkes / Signe de producteur	Process / Erschmelzungsart / Mode de fusion	Inspector's stamp / Zeichen des Sachverständigen / Poinçon de l'expert	Grade, Werkstoff, Nuance
W	E+AOD	DJ 2	Outokumpu 2205 UNS S32205 UNS S31803

Product, Erzeugnisform, Produit:
Stainless steel hot rolled, Plate, 1D/No.1, Cut edge

Line / Reihe / Ligne	Item / Position / Poste	Heat-Lot No / Schmelze-Lot Nr / Coulée n° - Lot No	Size / Abmessungen / Dimensions	Pieces / Stückzahl / Nombre	Quantity/unit, Menge/Einheit, Quantité/Unité
1	4	651761-479073:0	12.7 X 3048.0 X 6248.0 mm	6	

	Plate No, Blech Nr, Tôle No
1	6738 3901,-3902,-4001,-4002,-4101,-4102

Chemical composition – Chemische Zusammenzetzung – Composition chimique %
Heat – Schmelze – coulee No

	C	Si	Mn	P	S	Cr	Ni	Mo	N
Min						22.0	4.50	3.00	0.14
Max	0.030	1.00	2.00	0.030	0.020	23.0	6.50	3.50	0.20
651761	0.021	0.46	1.35	0.027	0.001	22.2	5.67	3.13	0.17

Distribution
Certificate via e-Services.

Direct. T = Transverse L = Longitudinal | Top/Bot F = Front B = Back | Form P = Flat C = Round | Location 1 = Center 2 = Close to Surface 3 = Standard 4 = At ¼ of thickness

Test results - Prüfergebnisse - Résultats des essais (1N/mm² = 1MPa)

Mechanical properties
Hardness Rockwell C ASTMA370/Tensile testing ASTMA370

Plate-no	Direct.	Top/Bot	Location	Form	Temp. °C	Rp 0,2 MPa	Rm MPa	A50 %	HRC HRC
Min						450	655	25	
Max									28
4101	T	F	2	P	20	556	785	44	21

ASTM A923 B Charpy impact test (V-notch)

Plate-no	Direct.	Top/Bot	Location	Form	Temp. °C	Value1 J	Value2 J	Value3 J	Average J	Lat ex1 mm	Lat ex2 mm	Lat ex3 mm
Min									54	0.38	0.38	0.38
4101	T	F	2	P	-40	167	152	131	150	1.65	1.53	1.40

Corrosion
ASTM A923 Method C 24h

Plate-no	Direct.	Top/Bot	Location	Form	Temp. °C	Cor.rate mdd
Max						10
4101	T	F	3	P	25	0

Metallographic testing
ASTM A923 Method A

Plate-no	Result
4101F	OK

The results comply with the Requirements of the order - Die gestellten Anforderungen sind erfüllt - Les résultats sont conformes aux exigences de la commande

Outokumpu Stainless AB
BA Quarto Plate,Quarto Plate Europe
SE-, 693 81 DEGERFORS
SWEDEN
Regoffice: SWEDEN
V.A.T no: SE556001874801
, Regno: 556001-8748

Computer generated original
Quality Inspector
Mathilda Berkeroth
Mathilda Berkeroth

HEAT # 651761
4101- 4102
TRACER # 0740276US

HEAT # 651761
3901-3902
TRACER # 0740274US

HEAT # 651761
4001-4002
TRACER # 0740275US

Receiving Inspection
By
Date 01/15/25

Job: 089-25
PO: 42820
Line: 002
HT: 651761

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Ferrite examination ASTM E562

Plate-no	Direct.	Top/Bot	Location	Form	Temp. °C	Ferrite %
4101	T	F	3	P	20	47

Ferrite examination Fischer Feritscope

Plate-no	Direct.	Top/Bot	Location	Form	Temp. °C	Ferrite %
Min						35
Max						65
4101	T	F	2	P	20	46

Other information
Country of melt: Sweden
Manufactured in Sweden.
Solution annealed at 1100°C and Quenched in water.
Dimensional control, visual inspection and grade check (PMI): Approved
Material conforms to NACE MR0175/ISO 15156 in supplied condition.
Material conforms to NACE MR0103/ISO 17945 in supplied condition.
Lateral expansion= min 0,38 mm (0.015 inch) according to ASME VIII Div.1 para UHA-51
DFARS Section 225.872-1 compliant
No weld repair allowed
Material delivered in accordance with Outokumpu document - Customer information concerning DIRECTIVES 2011/65/EU (RoHS), AND 2012/19/EU (WEEE), dated 30 January 2024
Material delivered in accordance with Outokumpu document - Control of Radioactive Contamination, dated 29 January 2024

Additional testing:
Ferrite examination ASTM E562: Ferrite (%)

Tolerance	Standard	Class
Flatness tolerance	ASTM A480	ASTM A480/A480M Plate
Length	ASTM A480	ASTM A480/A480M Plate
Thickness	ASTM A480	ASTM A480/A480M Plate
Width	ASTM A480	ASTM A480/A480M Plate

The results comply with the requirements of the order - Die gestellten Anforderungen sind erfüllt - Les résultats sont conformes aux exigencies de la commande

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